

with 4MB cache and 2.16GHz clock is able to outpace the PXIe-8130 with 1MB cache and 2.3GHz clock.

If your system has a latest processor, you can further enhance the productivity of your test system. By smartly utilising multi-core processors in the test system, the testing time can be reduced by running multiple threads for each task. Tasks such as frequency analysis are heavily dependent on the processor's computing performance for faster results. By optimising the algorithm to allocate each channel to multiple cores on the processor, you can improve the processor throughput. One of the reasons for the lower computing time is that the processor is able to execute Fast Fourier Transform (FFT) frequency analysis simultaneously on all cores.

Implement pipelining, improve throughput

Pipelining is a method used in a variety of fields to improve the throughput of an assembly-line-like system. It does not reduce the time to execute a test, but rather increases the number of tests that can be done simultaneously. More importantly, in pipelining, every test cycle takes as long to complete as the slowest cycle. But in most cases, the gain from implementing pipelining far outweighs the minor losses. Pipelining works best when each test takes almost the same time, as it minimises waste of time.

Pipelining works by delaying the start of testing each unit by one cycle. This delay allows later test sequences to run concurrently. Table I shows improvements in performance due to pipelining.

Re-order and schedule, maximise instrument utilisation

Once you have a faster processing and pipelining set-up, the next step is to improve instrument utilisation using test automation software. Even when you optimise your processor to improve upon the compute-intensive tasks, there is always a chance that

TABLE I
Performance Improvements Due to Pipelining

Standard Sequential Testing									
	Cycle 1	Cycle 2	Cycle 3	Cycle 4	Cycle 5	Cycle 6	Cycle 7	Cycle 8	Cycle 9
Unit 1	Test 1	Test 2	Test 3						
Unit 2				Test 1	Test 2	Test 3			
Unit 3							Test 1	Test 2	Test 3
Pipelining Implemented									
	Cycle 1	Cycle 2	Cycle 3	Cycle 4	Cycle 5	Cycle 6	Cycle 7	Cycle 8	Cycle 9
Unit 1	Test 1	Test 2	Test 3						
Unit 2		Test 1	Test 2	Test 3					
Unit 3			Test 1	Test 2	Test 3				

TABLE II
Performance Improvements Due to Scheduling and Pipelining

Standard Sequential Testing									
	Cycle 1	Cycle 2	Cycle 3	Cycle 4	Cycle 5	Cycle 6	Cycle 7	Cycle 8	Cycle 9
Unit 1	Test 1	Test 2	Test 3						
Unit 2				Test 1	Test 2	Test 3			
Unit 3							Test 1	Test 2	Test 3
Parallel Test with Scheduling									
	Cycle 1	Cycle 2	Cycle 3	Cycle 4	Cycle 5	Cycle 6	Cycle 7	Cycle 8	Cycle 9
Unit 1	Test 1	Test 2	Test 3						
Unit 2	Test 3	Test 1	Test 2						
Unit 3	Test 2	Test 3	Test 1						

your instruments will lie idle. To prevent such wastage of resources, auto-schedule tasks to run whenever the instrument is detected to be idling.

Not every set of tasks can be reordered. Move the tasks that can be reordered to fill in the idle instrument cycles. If you compare Table II and Fig. 2, you will notice that this methodology allows you to further enhance productivity by reducing the test process from five cycles to three.

"You've effectively reduced your test time by 66 per cent, and you're making the most of your instruments by cutting instrument downtime to zero. The bottom line is that by taking advantage of the advanced parallel testing capabilities of NI's software-defined automated test platform, you can use one set of instruments (that is, one test station) to test multiple devices in parallel. This, in turn, means you can test the same number of devices with fewer test stations, thereby drastically reducing the capital cost of test equipment without sacrificing test

throughput," explains Satish Mohanram, business development manager, National Instruments India.

Use composite measurements, reduce measurement time

In order to reduce measurement time, "The idea is to perform composite measurements instead of configuring each measurement separately. Because composite measurements calculate multiple measurement results on a single burst, they are more efficient than when each measurement is performed sequentially. Multicore processors enable this to be done in parallel, resulting in a faster overall test time when using a software-defined approach," suggests Tarun Gupta, business development manager at National Instruments.

Consider the case of measuring both power and error-vector magnitude (EVM) of a WLAN transmitter. Typical measurement times for these individual measurements are listed in Table III. In this case, each 802.11a/g